

Otsu's Thresholding Method

A classical statistical engine for autonomous image binarization — still essential in the age of deep learning.

COMPUTER VISION · CLASSICAL METHODS

The Core Challenge

Fixed Threshold

Hardcoded value (e.g., 128) – brittle, breaks under any lighting shift

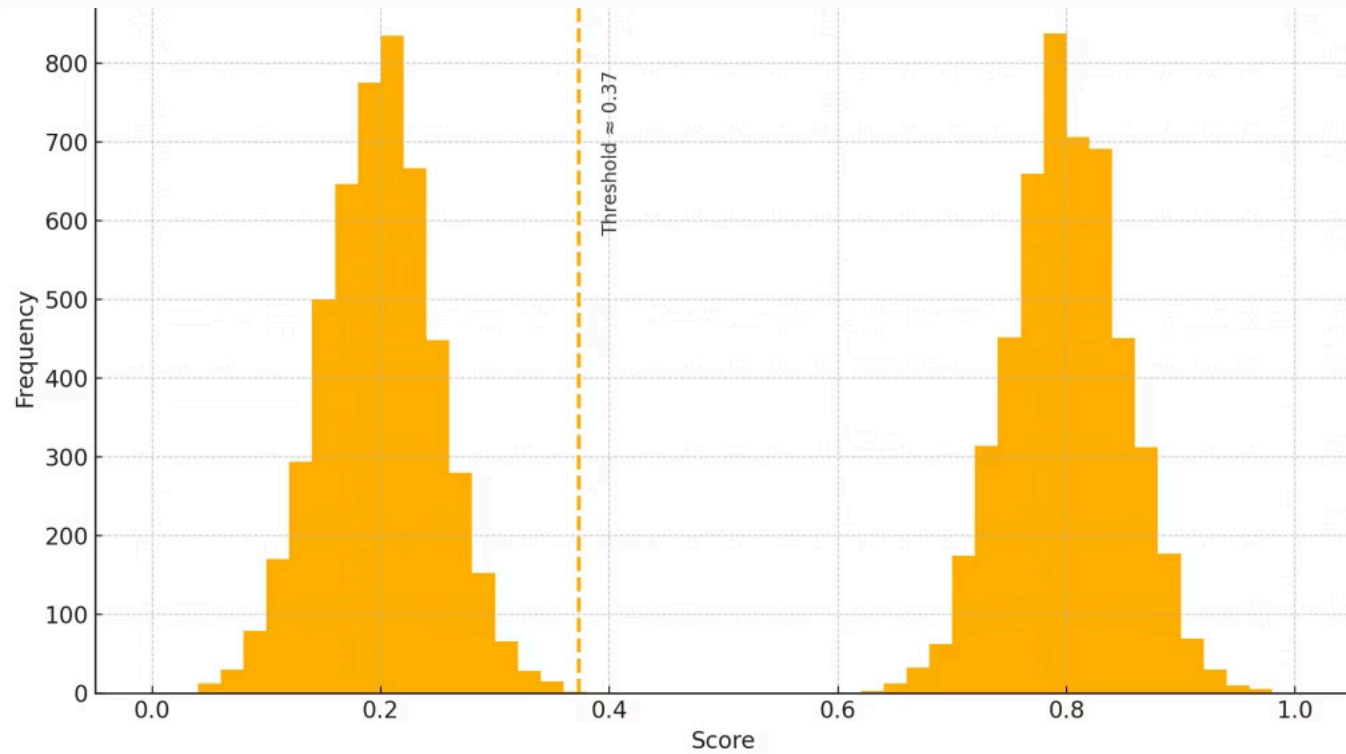
Real World

Clouds, exposure drift, sensor noise – intensity values are never stable

The Requirement

A **dynamic algorithm** that computes the optimal threshold **autonomously**, adaptively, and in real time – with zero manual calibration.

The Statistical Landscape



Otsu's Insight

Skip pixel-by-pixel spatial analysis. Instead, analyze the **intensity histogram** at a macro level.

A geometric problem becomes a **statistical one**: find the valley between two populations.

Peak 1 – Background

Cluster of dark pixel intensities

Peak 2 – Foreground

Cluster of bright pixel intensities

Step 1 – Probability Normalization

From Counts to Probabilities

$$p(i) = \frac{n_i}{N}$$

Where n_i = pixel count at intensity i , and N = total pixels.

Why It Matters

Produces a **Probability Mass Function** — making the algorithm **scale-invariant**.

Works identically on a 480p thumbnail and a 4K data stream.

Step 2 – Class Partitioning & Weights

For every candidate threshold t , the histogram is split into two classes:

C_B – Background

Intensities from 0 to t



C_F – Foreground

Intensities from $t + 1$ to 255

Class Weight Formula

$$\omega_0(t) = \sum_{i=0}^t p(i)$$

What it tells us

The **cumulative fraction** of the image that would be labeled as background at threshold t .

Step 3 – Class Centroids

Class Mean – Center of Gravity

$$\mu_0(t) = \frac{\sum_{i=0}^t i \cdot p(i)}{\omega_0(t)}$$

ⓘ Not the global image mean – each class has its own independent centroid.

As threshold t shifts, pixels switch classes and **both means move dynamically** across the intensity axis.

Step 4 – Between-Class Variance

$$\sigma_B^2(t) = \omega_0(t) \omega_1(t) [\mu_0(t) - \mu_1(t)]^2$$

The Optimization Criterion

Maximize $\sigma_B^2(t)$ – push the two class centroids as far apart as possible.

The Mathematical Shortcut

In a closed system, maximizing **between-class variance** is equivalent to minimizing **within-class variance**. Perfect separation guaranteed.

The Brute-Force Search Pipeline



Compute $p(i)$

Loop 256 t

Compute $\sigma^2_B(t)$

Select t^* (max)

Linear complexity $O(L)$ — executes in under **1 millisecond**. Ideal for real-time video streams.

When Otsu Fails

Global Algorithm Limitation

A single threshold applied to the **entire frame** assumes a clean bimodal histogram.

→ Gaussian Pre-filter

Smooths sensor noise — restores clean histogram peaks

→ Multi-Level Otsu

Tracks multiple thresholds for complex scenes

→ Adaptive Windowing (Sauvola)

Local block-level thresholds for non-uniform lighting

Real-World Case Study: Medical Imaging

Brain MRI Segmentation

Tumor tissue exhibits **distinct density and brightness** vs. healthy soft tissue — a natural bimodal distribution.

Automated Mask

Skull and background eliminated instantly

Zero Operator Bias

100% reproducible, no manual annotation

DL Handoff

Clean, preprocessed data fed to neural networks